

Question	Marking Guidance	Mark	Comments
10.1	1. Change in <u>DNA</u> base sequence/triplet; 2. Change in (sequence of) <u>amino acids</u> OR Change in primary/tertiary/3 ⁰ structure; 3. (Results in) rapid/uncontrollable cell division;	3 (3 x AO2)	2. Ignore reference to protein not being formed. 2. Reject (different) amino acids formed. 2. Ignore 3D structure. 3. Accept cell division cannot be regulated. 3. Ignore growth. 3. Accept cell replication but ignore cell reproduction.

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10.2	1. Use of PCR to amplify (DNA/sample); 2. Cut (DNA) using restriction endonuclease/enzymes; 3. Separate (DNA fragments) using electrophoresis; 4. Addition of (labelled) DNA probes and binding (by DNA hybridisation); 5. (Mutations) identified by fluorescence/radioactivity OR Compare positions/bands (to known) DNA sample with (all harmful) mutations;	4 max (3 x AO1, 1 x AO2)	1. Accept description of amplification. 3. Accept use of microarray for electrophoresis. 4. Ignore primers. 4. Reference to probe being complementary is insufficient. 5. Accept identification using X-ray/photographic/film/autoradiography or UV light. Note if only DNA sequencing is used award max 3 marks for the following. 1 Use of PCR to amplify (DNA/sample); 2. Sequence the DNA sample; 3. Compare DNA sequence with known DNA sequence of mutation;

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10.3	1. (Drug) binds to (oestrogen/ER) receptor; 2. Prevents binding of oestrogen/hormone; 3. No/fewer transcription factor(s) bind to <u>promoter</u> OR RNA polymerase not stimulated/activated;	3 (3 x AO2)	1. Accept (inactive) transcription factor for receptor. 1 and 2. Reject active site/enzyme - substrate complex once only.

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10.4	1. High/increased (concentration of) PSA not always linked to (prostate) cancer OR High/increased (concentration of) PSA could be a false positive; 2.(Could be) due to urinary infection OR (Could be) due to enlarged prostate;	2 (2 x AO2)	2. Accept 'urine infection'.

Question	Marking Guidance	Mark	Comments
10.5	1. (Drugs could) increase methylation of oncogene(s); 2. (Drugs could) decrease methylation of tumour suppressor gene(s); 3. (Increased) methylation of DNA/gene(s) inhibits transcription/expression (of genes) OR Decreased methylation of DNA/gene(s) stimulates transcription/expression (of genes); 4. Decreased acetylation of histones inhibits transcription/expression (of genes) OR (Increased) acetylation of histones stimulates transcription/expression (of genes);	3 max (1 x AO1, 2 x AO2)	Ignore methylation of histones and acetylation of DNA/genes. Ignore proto-oncogenes. 3. Accept promoter (region) for DNA/gene 3 and 4. Ignore 'switching on' and 'switching off' genes once but accept as alternative(s) for 1 mark if used correctly in context of transcription/ expression for both points 3 and 4.